Lab 2 — Transforming Space

Friday, October 23 (~20 min)

Last updated: October 12, 2026.

In class Friday, October 23 (same day as Lecture 26 / Quiz 7). Quiz 7 is Week 8 systems — **not** this lab.

Credit: Lab points are for effort in class, not for a Canvas file upload. Preview the one-pager; write on the hard copy handed out in class.

Learning objectives

1. Build a 2×2 matrix and compute Ax.
2. Predict and check a geometric map (scale / reflect / rotate).
3. Apply one matrix to several points stored as columns.

Math connections

Matrices and Ax (Lectures 18–20).

Key syntax (this lab)

Task	Octave	Python
Matrix	`A = [2 0; 0 1/2]`	`A = np.array([[2.,0.],[0.,0.5]])`
Matrix–vector	`A*x`	`A @ x`
Many points as columns	`Y = A*X`	`Y = A @ X`
Dimensions	`size(A)`	`A.shape`

Ops strip

C2.0. Enter $A = \text{diag}(2, 1/2)$ and $x = (1, 1)$. Print `size(A)` / `A.shape` and compute Ax.

Expected: A is 2×2; $Ax = (2, 0.5)$.

Application

Triangle columns $X = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

C2.1. Predict AX for $A = \text{diag}(2, 1/2)$ without software, then compute. Geometric effect in one sentence?

Expected: $AX = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 0.5 \end{bmatrix}$; stretch x_1 by 2, shrink x_2 by 1/2.

C2.2. Reflection through the x_1-axis: $R = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Compute Rx for $x = (2, 3)$ and RX.

Expected: $Rx = (2, -3)$; $RX = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$.

C2.3. Rotation 90° CCW: $Q = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Where does $(1, 0)$ go? Check $Q \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Expected: $(1, 0) \mapsto (0, 1)$.

One-pager

Preview the Lab 2 ticket; write on the hard copy in class.